Nader Khalil

AI Application Engineer

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PROFESSIONAL SUMMARY

AI application engineer with a B.Sc. in Software Engineering, building reliable LLM applications, RAG pipelines, and agentic workflows. Specialized in vector search (pgvector), autonomous tool calling with human review, automated AI safety evaluation suites (prompt injection, scope gating), and robust full-stack integration across TypeScript, Python, and PostgreSQL.

TECHNICAL SKILLS

AI & LLM Systems: RAG architectures, pgvector, tool/function calling, human review, prompt caching, Vercel AI SDK, LangChain

Full-Stack & APIs: Next.js, React 19, TypeScript, Python (FastAPI), REST APIs, layered architecture, Auth.js

Testing & CI/CD: Playwright (e2e), Vitest, automated LLM eval suites, PHPUnit, race tests

AI Safety & Evals: Automated prompt injection testing, scope gating, structured output parsing (Zod), citation grounding, rate limits

Data & Infrastructure: PostgreSQL + pgvector, MariaDB, Prisma, row locks, Docker, GitHub Actions CI/CD

Languages: Python, TypeScript, JavaScript, SQL, C, C++, Bash, Linux

TECHNICAL PROJECTS

AI Booking Platform 2026

Live build: booking.naderkhalil.dev · Case study: naderkhalil.dev/case-study.html

- Engineered an autonomous RAG assistant answering user questions with citations from admin-managed docs and role-filtered vector retrieval. (Next.js 16, TypeScript, PostgreSQL + pgvector, Prisma, Vercel AI SDK).
- Designed tool calling with Human-in-the-Loop review: the AI prepares booking and cancellation proposals as interactive cards that execute mutations only after explicit user confirmation.
- Built an automated AI evaluation suite with 120 live checks gating prompt injection, data leaks, and scope drift; verified concurrency safety with row locks under 766 automated tests in CI.

AI Appointment Scheduler 2025

Live build: appointment.naderkhalil.dev

- Built a conversational booking assistant using Python (FastAPI), LangChain, and Next.js, parsing natural-language scheduling requests into structured API payloads for calendar availability checks and slot reservations.

Student-Faculty Services v2, *Al Ain University capstone rewrite* 2026

Live build: student-faculty-services-v2.infinityfree.io · Case study: naderkhalil.dev/case-study-v2.html

- Architected a high-reliability framework-free PHP 8.4 REST API (46 endpoints) and React 19 frontend; verified 0 double bookings across 8-student concurrent race tests with 500+ tests and 100/100 accessibility.

PROFESSIONAL EXPERIENCE

AI Trainer & Mentor, *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, prompt structuring, and autonomous tool calling for the accredited AgenticVerse program; mentored teams building AI prototypes for ADNOC, FANR, and Khalifa Fund.
- Guided participants through API schema validation, structured output parsing, and debugging in Python and TypeScript.

QA & Software Tester, *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on an internal student platform using black-box scenarios; authored reproduction reports with logs and steps to help developers isolate defects quickly.
- Tested features across browsers and viewports; re-tested defect fixes across user flows to ensure zero regressions before release.

IT Support & Technical Staff, *Ticketmaster* 12/2025

- Provisioned and deployed entry ticketing hardware and access scanners for a high-volume international event; maintained 100% operational uptime through real-time hardware health checks.
- Served as on-site technical point of contact, delivering rapid incident resolution and verifying network connectivity under strict SLAs.

Software Engineering Intern, *Timex IT Consulting* 01/2024 – 04/2024

- Developed C# MinimalAPI CRUD endpoints with request validation and standardized HTTP response handling.
- Supported software release rollouts by identifying, documenting, and verifying functional defect fixes on client accounting software.

EDUCATION & CERTIFICATIONS

B.Sc. in Software Engineering, *Al Ain University*

Computer Science Diploma, *42 Abu Dhabi*

MOEI × 42 Abu Dhabi AgentEra Hackathon, *Honoree*

CyberQ CTF, Top 10 Finalist, *UAE Cybersecurity Council*

Hack The Box University CTF, *Top 12% (119/1,014 teams)*

RecognAIze Hackathon, *AI on Watch with Datamind*

Languages: Arabic (Native), English (Professional Proficiency)

Portfolio & Case Studies: naderkhalil.dev